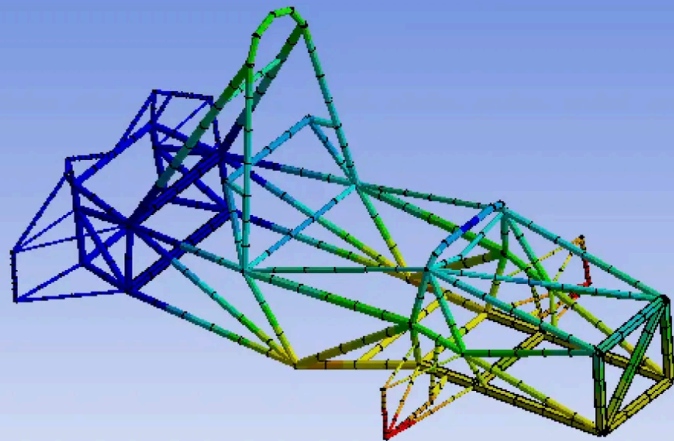
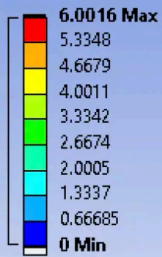


Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s
12/28/2024 11:08 PM

ANSYS
2023 R2

OpenGL < 4.5



SN4 Simulations

➤ Subteam

 Chassis Structures

👤 Project Manager

Empty

👤 Team

 Oliver Hsu-Che Chang

📌 Status

Active

☰ Program

SN4

✓ 5 more properties

Comments



Add a comment...

Simulations in both ANSYS and Solidworks inform rule compliance and assess the Factor of Safety under critical situations such as crash, cornering, and braking scenarios. Because it is impractical to build and do physical testing on many configurations of parameters, using simulations is a cost-effective and efficient way to iterate before settling on a final design.

The connection between simulations and spaceframe design helps aligns our engineering process with first principles, using analytical reasoning to iterate between chassis design.

This is our team's first year using ANSYS to do structural simulations on the car. Current progress has been done to solve for various conditions but there is a lot of opportunity in how to use this data and towards optimizing for performance.

Design Objections

- Optimize Torsional Stiffness
- Reduce Spaceframe Weight
- Improve Handling Performance

Types of Simulations

- Torsional Stiffness Analysis
- Modal Analysis (Natural Frequency Study)
- False Static Crash/Impact Analysis (Frontal, Side, Rear)
- Lateral Load Transfer Distribution (LLTD) Study
- Braking Force Analysis
- One-Wheel Bump with Simultaneous Cornering Load

(Source: <https://www.designjudges.com/articles/tube-frame-analysis>)

Methodology

This was the first year we used ANSYS for chassis simulations, allowing us to perform more advanced and accurate analyses. Why ANSYS?

- **Industry-Standard FEA:** ANSYS offers a more robust finite element analysis platform, enabling us to perform complex simulations with higher accuracy
- **Beam Models:** ANSYS allows for the creation of beam models, which represent the chassis structure using simplified line elements. This is a key factor to reduce the computational requirements while preserving accuracy for evaluating structural stiffness and load paths.
- **Post-Processing Tools:** ANSYS provided detailed visualization and data extraction tools, enabling a deeper understanding of stress distributions and deformation patterns.

The methodology outlined below details the tools, setup, and process used to analyze the chassis design. In general, the process starts with ANSYS Spaceclaim to clean up the model, then ANSYS Mechanical for specifying load cases and constraints.

Tutorials

 [Formula Racing at UC Davis \(FRUCD\) FSAE Torsional Stiffness Simulation in ANSYS - Part 1](#)

 [Google Docs FEB | Chassis | Torsion Simulations Guide Document](#)

The guide document is still in process, but has helpful screenshots of step-by-step instructions on the process of setting up a simulation.

Set Up

1. Exporting from Solidworks

Export the chassis and suspension assembly as a PARASOLID file type (.x_t).

Why Parasolid File Type? Format developed by Siemens that focuses on geometric and topological data for lightweight transfer, universal to most simulation software that require realistic force application and load paths

Afterwards, open up ANSYS SpaceClaim to edit the model.

2. Convert to Beam Model

After importing the parasolid file to ANSYS, group the chassis beams together and use the extract tool to extract the centerlines of the beams.

Use the connect tool to bring all the joints together. A helpful tip that I found was starting at a distance then gradually decreasing the distance until there are none remaining.

Tip! I came across an error later in the process where the software struggles because curved beams in the front roll hoop. It may be easier to simplify the model by getting rid of the curved tubes and representing the connection as between straight lines.

The split tool is then used to create points at where the suspension parts will later connect to the spaceframe.

3. Model Suspension

With the sketch command, place points at the points where the a-arms connect to the upright, as well as for the rocker to spring and rocket to pushrod connections. Do this for one side as we will then mirror to the other side.

Afterwards, use lines to connect the points. Extract from the parasolid model the profiles of the pushrod and a-arms. These profiles will be used to convert the lines into beams by selecting on the beams then assigning the right profile to each.

4. Share Vertices

The Share Tool in ANSYS SpaceClaim ensures that connected components in a model share common edges or faces, creating seamless geometric interfaces between parts. The tool helps eliminate gaps or overlaps and groups the parts for ANSYS Mechanical.

After this step, we move on to ANSYS Mechanical to set up the simulation!

Results

Through ANSYS Mechanical, boundary conditions are inputted, materials are specified (4130 Steel), and the model is meshed. The specific loads and boundary conditions are different depending on the specific case simulated, but follow similar patterns.

For a full slidedeck of all simulations run on past models of the spaceframe (starting with SN3 as a baseline), visit the following link:

FEB | Chassis | Spaceframe Analysis Slidedeck

Chassis | Frame Analysis



<https://docs.google.com/presentation/d/117f-W2-C9EZ...>

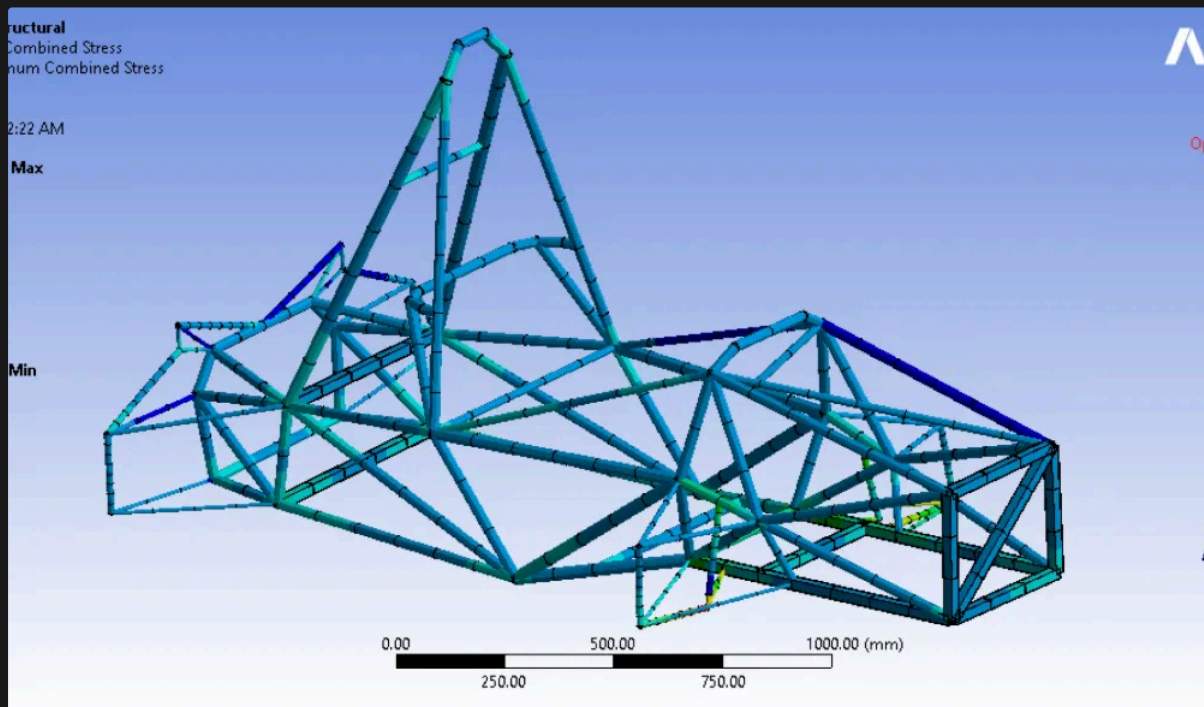


Chassis | Frame Analysis

The following results are the results of November 2024 simulations, roughly during the first SES submission.

Torsional Stiffness

One of the main parameter that was found is the spaceframe's torsional stiffness. This measure of the car's rigidity relates to the handling performance and car's response to heavy loads. Optimizing the torsional stiffness against other factors such as weight is important to improving the car's performance at events.



Boundary Constraints

- Rear uprights are fixed, which mirrors the testing jig setup
- Front suspension rotation constraints, constraint to just rotation on x-axis
- Simulate suspension with infinite stiffness to focus on effect on car

Forces

- 1800N up on inside front upright, 1800N down on opposite front upright

Forces were calculated by the max lateral acceleration and by including a 1.5 FOS into the calculation.

Lateral force experienced during cornering

$$\text{Force}_{\text{Lateral}} = \text{mass} * \text{acceleration}_{\text{Lateral}}$$

Distribution of Lateral Forces

$$\text{Force}_{\text{Lateral, Loaded Side Wheel}} = F_{\text{Lateral}} * 0.5$$

Torque Calculation

$$\text{Torque} = \text{Force}_{\text{Lateral, Loaded Side Wheel}} * t * 0.5$$

Factor of Safety Multiplier

$$\text{Torque}_{\text{Sim}} = \text{Torque} * \text{FOS}$$

Constants

$$m = 250 \text{ [kg]}$$

$$a_{\text{lat}} = 1.5 \text{ [G]}$$

$$t = 1.2 \text{ [m]}$$

$$\text{FOS} = 1.5x$$

Results

$$\text{Force}_{1.5 \text{ FOS}} = 1839$$

$$\text{Torque}_{1.5 \text{ FOS}} = 1699$$

Calculations

Torsional stiffness is calculated by placing two displacement probes at the nodes where the a-arms connect to the chassis and measuring the vertical displacement upwards. Using the equation below, we can find the torsional stiffness by dividing the torque enacted on the car by the angle of deformation. The final result is how much the car torques for a every 1 degree increase in deformation.

$$\text{Torsional Stiffness } (C_T) = \text{Torque } (T) / \text{Angle of Deformation } (\alpha) [\text{Nm/deg}]$$

$$\alpha = \arctan((\mu_1 + \mu_2) / L)$$

$$T : 0.557\text{m} * 3600\text{N} = 2005.2 \text{ Nm}$$

$$\alpha : \text{atan}((\mu_1 + \mu_2) / L) = \text{atan}(8.0284\text{mm} + 9.0895\text{mm}) / 380\text{mm}$$

$$1325.74447254 \text{ Nm/deg}$$

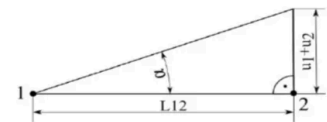


Figure 4: Deformation triangle.

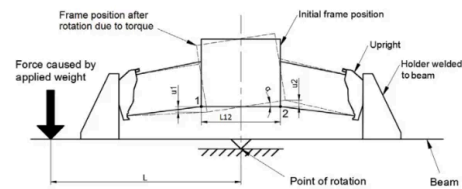


Figure 3: Simplified front view of rotating beam.

Results:

- $CT = 1325.74447254 \text{ Nm/deg}$
- $(6.7157 \text{ mm RHS} / 3.3639 \text{ mm LHS})$
- Displacement: 32.928 mm

Modal Analysis

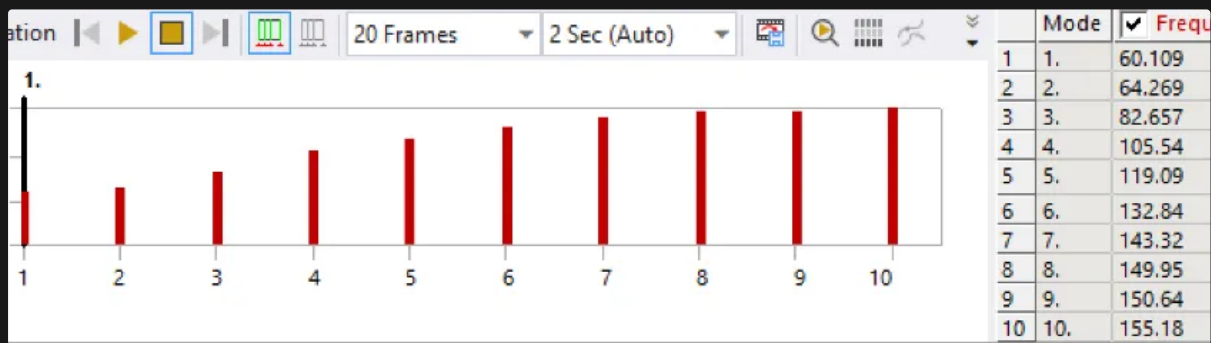
We can use modal analysis to determine the natural frequencies and mode shapes of the spaceframe. This can be used to ensure that there's no resonance overlap that can lead to failure,

Natural frequencies: The vibration frequencies at which a structure tends to oscillate when disturbed, while mode shapes describe the deformation patterns of the structure at those frequencies.

For the simulations, the different modes identify how the chassis deforms at specific frequencies.

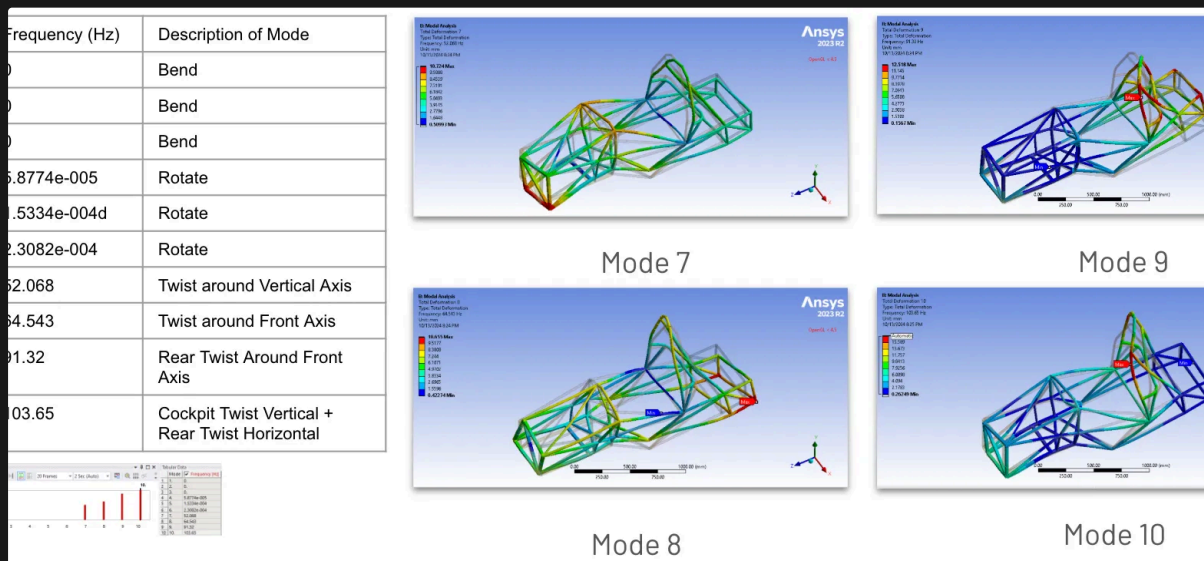
Method 1: Constrained

- **Boundary Conditions:** 4 Fixed Support Points (Primary load transfer points)



Method 2

- **Boundary Conditions: None**



We found these results to generally align with other teams. Further research is needed on how to use the modal analysis data towards improving functionality and impact on the chassis, as well as how to experimentally validate this data. Because of our limited time and resources, we choose to prioritize the other simulations.

Cornering

We specifically focus on steady-state cornering: the instance when the car rounds a turn at a constant speed and radius of curvature. We can look at the areas of stress concentration and assess the FOS to ensure safety, even during the heaviest cornering conditions.



Boundary Conditions:

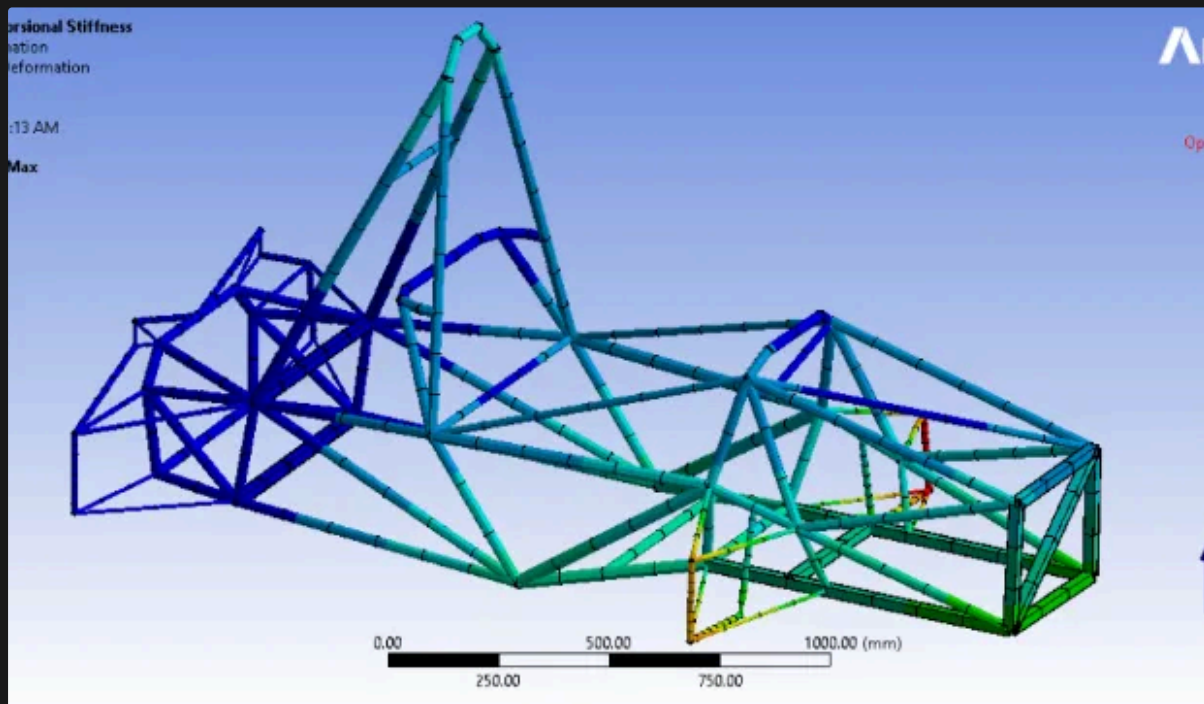
- Rigid Suspension, Fixed inside (LHS) uprights, forces of 3845.54 N lateral, no vertical forces

Results

- Displacement: 10.989 mm
- Max Combined Stress: 182.34 MPa
- FOS: 2.4

Cornering + Bump

A cornering + bump simulation is when both lateral forces (cornering) and vertical forces (bump) are applied. As compared to just cornering, this case better represents real-world driving. Imagine that the car is taking a sharp curve over an uneven surface or pothole/curb.



Boundary Condition:

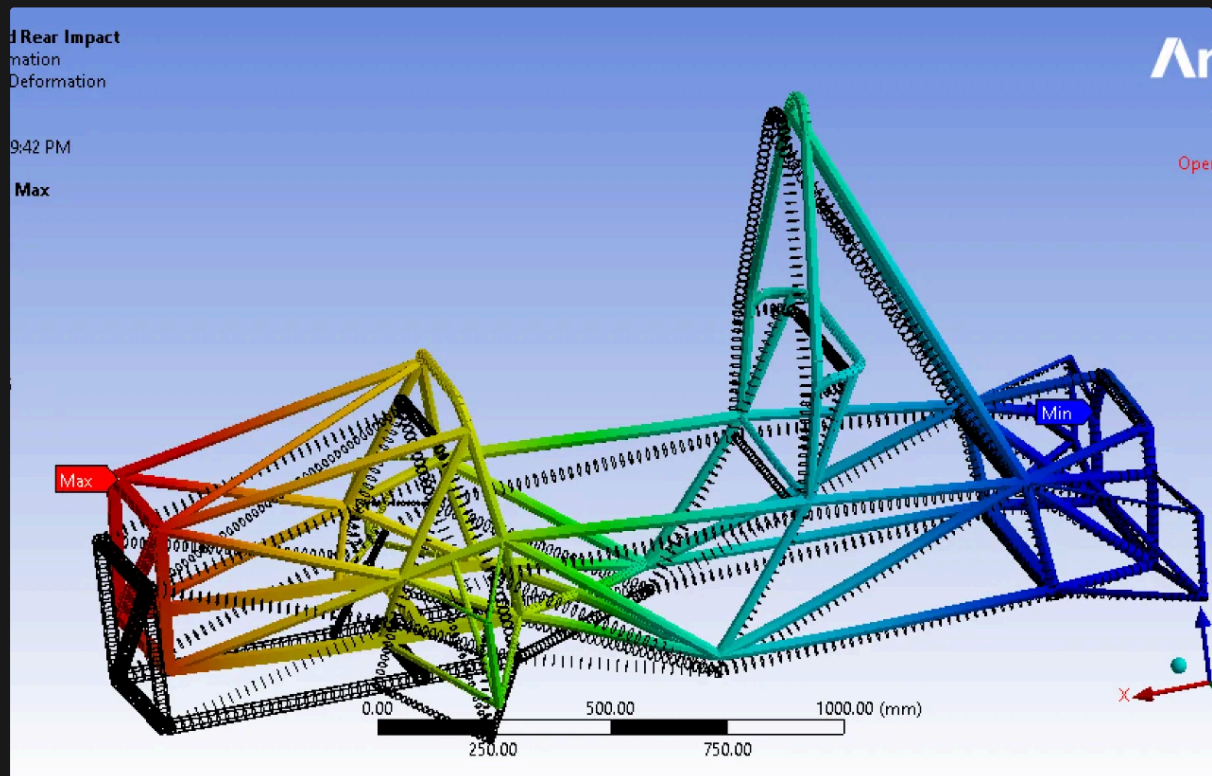
- Rigid Suspension, Fixed inside (LHS) uprights, forces of 3845.54 N lateral, rear outside 1510.20 N vertical, front outside 1478.59 N vertical

Results:

- Max Stress: 111.73 Mpa
- FOS: 4.11

Front Impact

The front impact simulates the forces experienced by the chassis during a frontal collision. We get the force by assuming the max acceleration forward by the car. Using this information, we can understand how the impact translates throughout the car. The chassis must be designed to maintain structural integrity, protect critical components, and minimize deformation to ensure driver safety in the event of a crash.



Boundary Conditions:

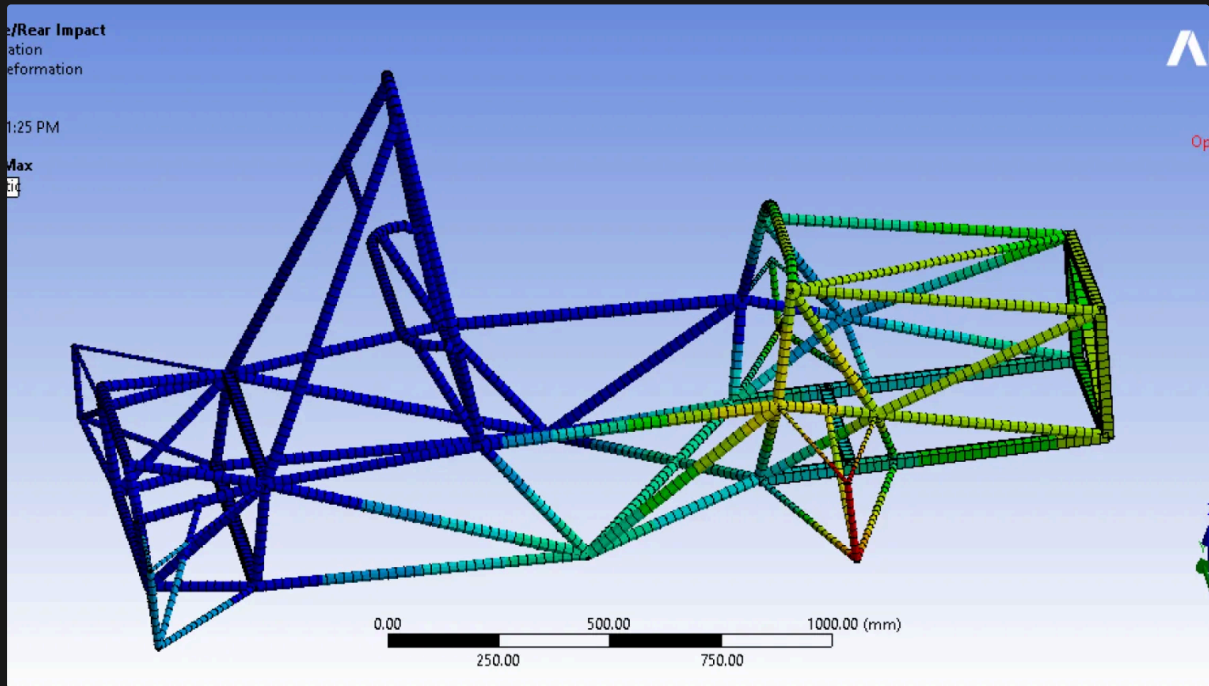
- 20G Front Crash (~60kN Force) & Fixed Rear Nodes

Results:

- Max Displacement: 8.738 mm
- Max Stress: 79.302 MPa
- Max Combined Stress: 353.32 MPa
- FOS: 1.231

Side Impact

For a side impact, the same results are assessed but during a lateral collision.



Boundary Conditions:

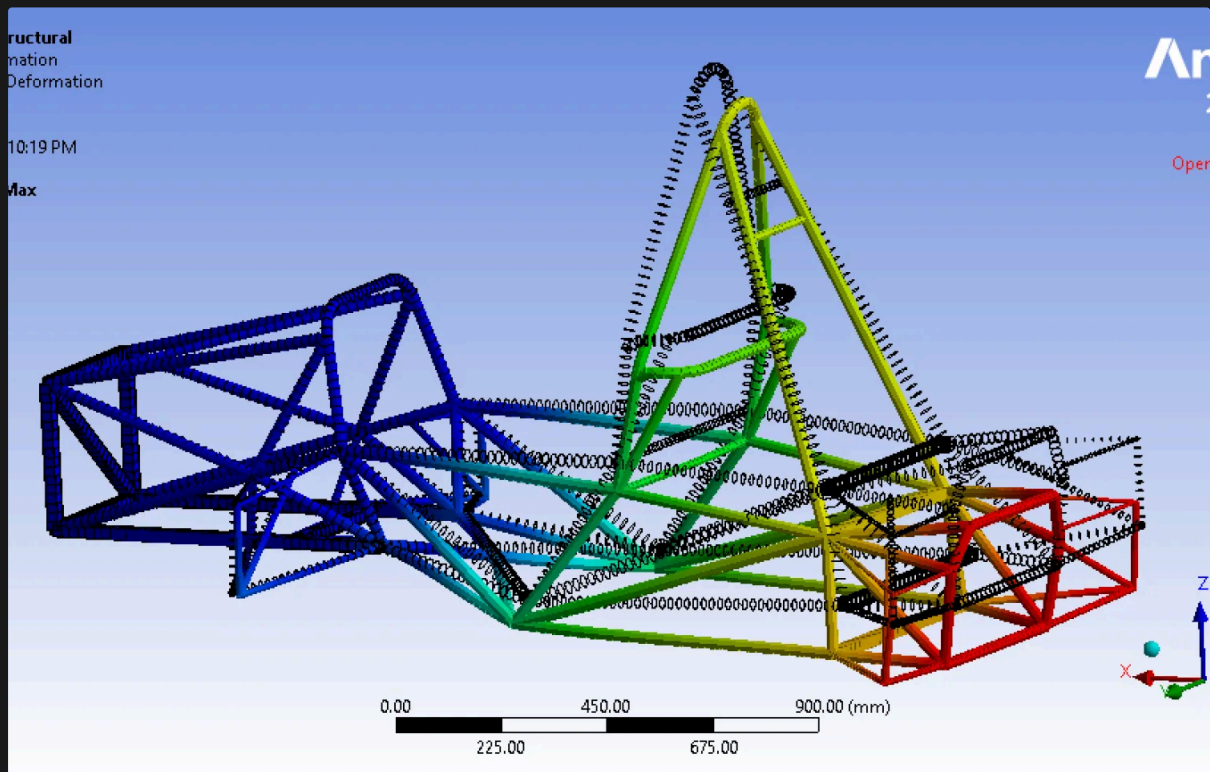
- 2.5G Side Crash (~7000N Force)
- Fixed Side Nodes

Results:

- Max Displacement: 2.4254mm
- Max Stress: 12.806 MPa
- Max Combined Stress: 142.17 MPa
- FOS: 3.059

Rear Impact

The rear impact simulation ensures that the chassis can protect the drivetrain, suspension mounts, and other critical components while minimizing deformation to maintain safety and performance.



Boundary Conditions:

- 10G Rear Crash (~30kN Force)
- Fixed Front Nodes

Results:

- Max Displacement: 9.225 mm
- Max Stress: 25.078 MPa
- Max Combined Stress: 279.05 MPa
- FOS: 1.559

Takeaways:

- Rear impact should be good for >15 g of acceleration given fos of ~1
- Generally rear crashes are around 10-15 g of deceleration ([Formula SAE Impact Attenuator Testing & Validation](#))
- Main stress concentration happens at the cross below the driver for a rear impact
- Unrelated, but monocoque teams do drop test for physical validation of simulations ([Video](#))



Key Lessons Learned & Past Problems

- Use normalized 4130 compared to annealed. Normalized steel is produced by heating the steel to very high temperatures then allowing it to cool.
- For torsional stiffness testing, the pushrods and springs should be rigid (set springs to infinite stiffness or use beams in place of them)
- Use the following joints for the proper constraints of DOF during a torsional stiffness sim: general instead of revolute to restrict rocker DOF & Spherical for A-Arm to Chassis/Upright to A-Arms

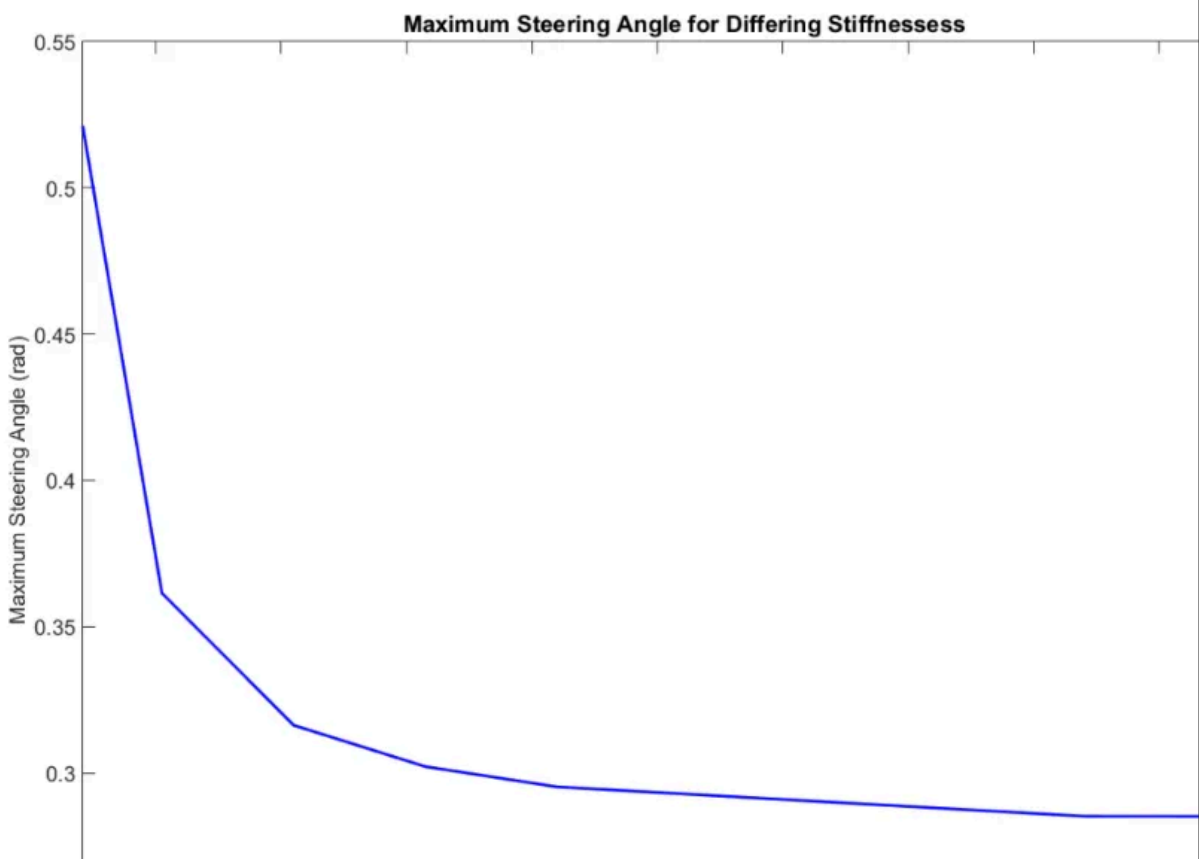
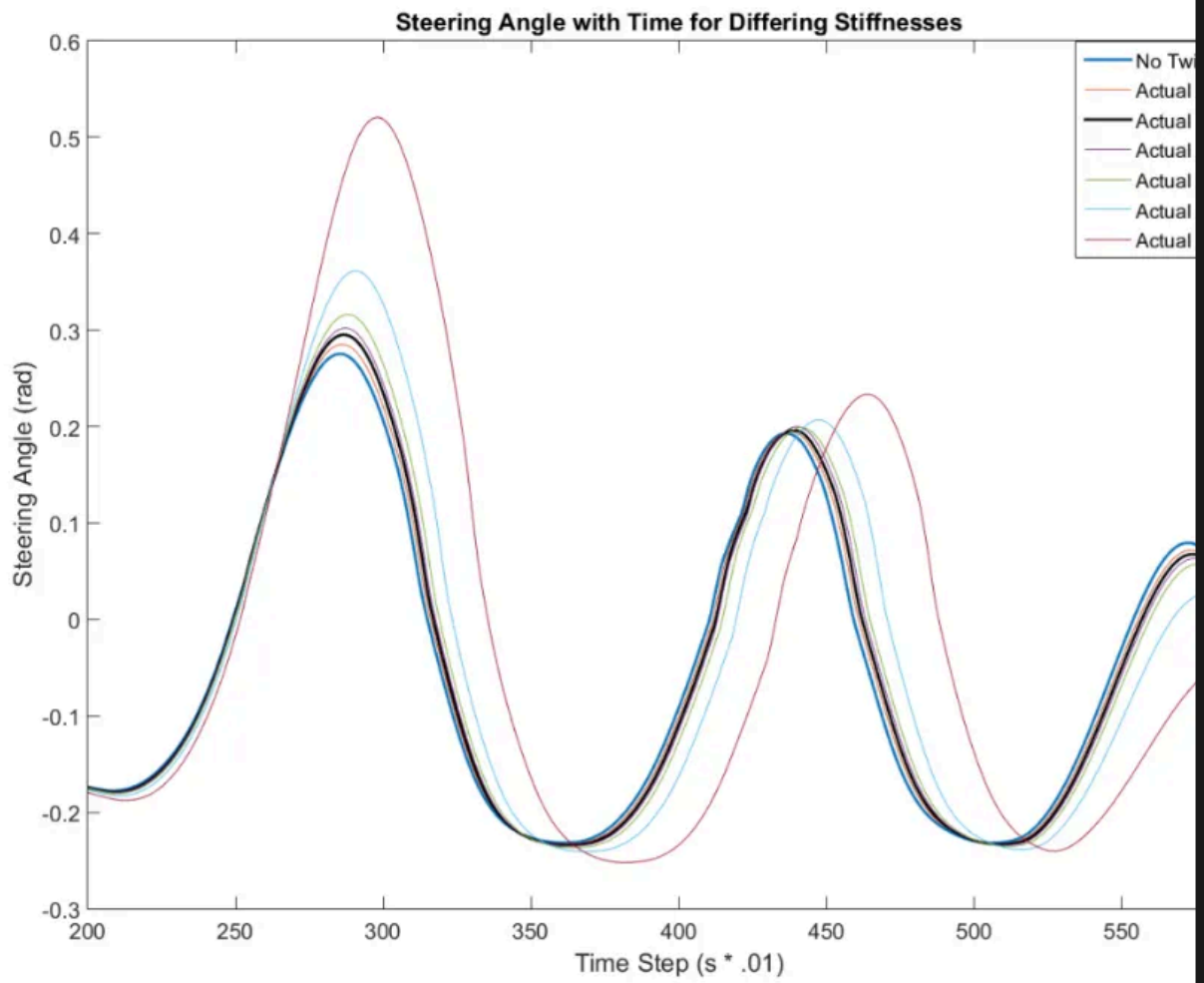
Target Torsional Stiffness

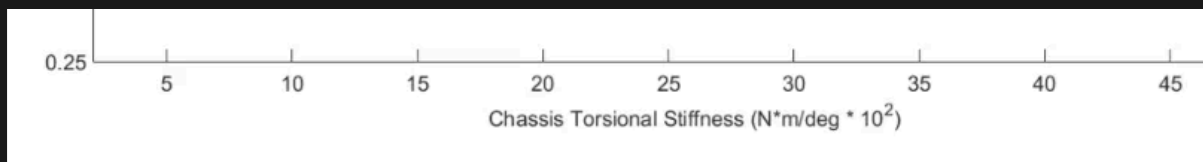
One of the next steps in the simulation process is understanding how to set goals for torsional stiffness. Currently, the simulations are able to get the value for any designed chassis. But what can this be judged against?

Currently, we've looked at existing research papers that help to translate the effects of torsional stiffness to handling performance

Steering Angle

Torsional stiffness is important for vehicle handling, especially during turns. A stiffer chassis improves drivability by reducing steering angle in transient conditions.





Overview

- During steady state turning, goal is to maintain handling /ease of driving while having lowest stiffness possible to minimize mass of chassis
- During transient period entering/exiting, different amounts of handling is required depending on load distribution, affecting steering radius (ex. Inc to Front → Understeer and larger turn radius required)
- Uneven total lateral load transfer distribution is inevitable can be tuned using roll stiffness, however:

"Stiffer chassis minimizes the need for differences in front-to-rear roll stiffness to maintain a consistent handling balance. The LLTD can only be controlled however, if the chassis is stiff enough to transmit the torques"

- Sims of different steer angles and speeds used to find optimal torsion stiffness

Results

- Steady State: Roll rates tune out torsional stiffness, which has little effect
- Transient: Stiffness reduces steering angle required for turns, making the car more difficult to drive
- Range of **1500+ Nm/deg** (Range due to manufacturing error from designed val)

(Study: MRacing)

Conclusion

Currently in the simulation process, we're able to quantify these metrics of FOS and torsional stiffness, but there's a gap in understanding what the goal and target values should be for each. Future research into how to set these goals and better understanding the relationship with dynamics performance is important for using this new data to qualify the extent of the car's performance and it's potential/theoretical peak.

Because simulations was started mid-development cycle, there's opportunity for future car designs to utilize simulations more across the entire design cycle. Having more iterations and "what if" can be weighed through simulation, helping validate hand-calcs and helping make more analytical decisions.

Next Steps

1. Parametric Study for Optimization

It seems that there are systems of varying systematically varying the torsional stiffness of the chassis to study its impact on performance metrics like suspension effectiveness and vehicle handling.

Step-By-Step

1. Prepare the chassis in ANSYS Spaceclaim
2. Define stiffness parameters (tube thickness, materials) and parametrize them.
3. Set up a parameter sweep with varying stiffness values.
4. Run simulations with applied torque, measuring angular deflection for stiffness.
5. Analyze results by plotting stiffness vs. weight and suspension performance.
6. Based on these results, we can determine the required stiffness that optimizes the other factors such as weight

2. Lap Time Simulations

Another way that's worth investigating is by using virtual car simulation softwares and see how various torsional stiffness values would effect lap times and handling performance. We can use this method in tandem with the other method to validate one another

Step-By-Step

1. Use software like **OptimumLap**, **VI-CarRealTime**, or a custom MATLAB/Simulink model.
2. Input the car's dynamics parameters
3. Set up a pre-defined FSAE track or create one that replicates competition layouts (e.g., skidpad, autocross)..
4. Run simulations with varying chassis torsional stiffness values,